

SmartClinic Management System

1. Project Overview

The **SmartClinic Management System** is a healthcare platform designed to streamline clinic operations and healthcare delivery through AI-integrated prescriptions, specialized accessibility (Elder Mode), and an intelligent 30-minute interval appointment scheduling system.

2. User Roles & Permissions

The system categorization ensures data security and role-specific workflows:

Role	Description	Key Permissions
Admin	Clinic managers/owners.	User management, system configuration, full access to all schedules, logs, and financial records.
Doctor	Medical professionals.	Manage patient records, view personal schedules, and utilize AI-assisted prescription generation.
Patient	Service recipients.	Profile management, appointment booking, and viewing personal medical history.

3. Core System Functionalities

User & Data Management

- **CRUD Operations** – Create, Read, Update, and soft delete profiles for Admins, Doctors, and Patients.
 - **Medical Record Keeping** – Digital storage of patient vitals, history, consultation notes, and lab reports.
 - **Financial Tracking (Proposed)** – Real-time invoice status updates:
 - Pending
 - Partial
 - Paid
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Appointment Lifecycle

Every appointment progresses through the following states:

1. **Scheduled**
 2. **Checked-in**
 3. **Completed / Cancelled**
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4. Intelligent Scheduling System

The system operates on a **30-minute interval appointment grid** with special logic for urgent cases.

Global Emergency Slots

- **11:00 AM** and **3:00 PM** are reserved for emergencies.
- These slots remain **hidden from standard patient views**.

Urgent Symptom Trigger

If a patient selects “**Urgent Symptom**” in the intake form:

- Emergency slots automatically become visible.

Privileged Access

- **Doctors and Admins** always see these emergency slots.
- They can **manually assign them** when required.

5. AI-Assisted Prescription Module

This module is available inside the **Doctor's Consultation Dashboard**.

Workflow

1. Doctor enters a diagnosis (example: *Type 2 Diabetes*).
2. Doctor clicks "**AI Suggest**".
3. System generates medication and dosage recommendations.

Clinical Oversight

- AI suggestions are **fully editable**.
- Doctor must click "**Finalize Prescription**" to save the record.

Audit Logging

The system automatically logs whether the prescription was:

- **AI-suggested**
 - **Manually entered**
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6. Accessibility Feature: Elder Mode

A global toggle in the **Patient Portal** designed for users with visual or cognitive challenges.

UI/UX Specifications

Typography

- Minimum font size: **22pt**

Visuals

- High contrast theme
- **Pure Black text** on **Pure White background**

Interactions

- Minimum button size: **80 × 80 px**

Simplified Appointment Booking (Wizard)

Instead of a calendar grid, the system provides a **step-by-step interface**.

Screen 1 – Select Date

Screen 2 – Select Time Slot

Screen 3 – Enter Reason for Visit

7. Notification & Reminder System

Automated notifications ensure appointment efficiency and reduce no-shows.

Patient Reminders

- Sent **24 hours before appointment**
- Sent **2 hours before appointment**

Urgent Notifications

- Instant alerts sent to **Doctors and Admins** when an urgent slot is booked.

Status Updates

Patients receive notifications when:

- Lab reports are uploaded
 - Payments are confirmed
 - Urgent bookings are confirmed
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8. Admin Module

Dashboard & Analytics

- Real-time clinic KPIs (appointments, revenue, no-shows, occupancy rate)
- Custom reports (daily, weekly, monthly, yearly)
- Export reports as **PDF or Excel**
- Heatmaps of peak booking hours

User Management

- Create, edit, and deactivate Doctor/Patient profiles
- Assign roles and permissions
- Reset passwords and manage login sessions
- Bulk import/export records

Doctor & Staff Management

- Add or remove doctors and support staff
- Configure doctor working hours
- Assign specializations and departments
- Monitor doctor performance metrics

Schedule & Slot Management

- Manage the **30-minute appointment grid**
- Override emergency slots
- Block slots for holidays or maintenance
- Reassign appointments when a doctor is unavailable

Financial Management

- View invoices: **Pending / Partial / Paid / Overdue**
- Issue refunds or adjustments
- Configure payment gateways
- Generate financial summaries and tax reports

Notification & Communication

- Broadcast announcements to doctors or patients
- Configure notification templates (SMS, email, in-app)
- Monitor delivery logs and failure reports

Audit & Compliance

- View system audit trail
- Track AI-generated vs manual prescriptions
- Detect suspicious activities
- Manage data retention and compliance policies

System Configuration

- Configure Elder Mode defaults

- Manage clinic branding
 - Define cancellation and rescheduling policies
 - Configure automated reminders
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9. Doctor Module

Personal Dashboard

- View today's appointments
- Access weekly/monthly schedule
- Receive alerts for urgent bookings
- Monitor personal performance metrics

Appointment Management

- Accept or decline appointment requests
- Update appointment status:
 - Checked-in
 - Completed
 - No-Show
- Add internal notes

Patient Records

- View patient medical history
- Update consultation notes
- Upload lab reports
- Flag patients for follow-ups

AI-Assisted Prescription

- Enter diagnosis
- Generate AI medication suggestions
- Review and edit prescriptions
- Finalize and store prescriptions

Scheduling & Availability

- Set personal availability
- Block personal time-off
- Fill emergency slots

- Request schedule changes

Communication

- Receive real-time booking alerts
 - Send follow-up instructions to patients
 - Receive lab report alerts
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10. Patient Module

Profile & Account

- Manage personal profile
- Upload insurance/ID documents
- Toggle Elder Mode
- Configure notification preferences

Appointment Booking

- Search doctors by specialization
 - Book appointments using the **30-minute grid**
 - Use **Urgent Symptom option** to unlock emergency slots
 - Reschedule or cancel appointments
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Elder Mode Booking Wizard

Step 1: Select visit date

Step 2: Select time slot

Step 3: Enter reason for visit

Medical Records

- View consultation history
 - Access lab reports
 - Download or share records
 - Track vitals history
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Payments & Billing

- View invoices
 - Make full or partial payments
 - View payment history
 - Download receipts
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Notifications

- Appointment reminders
 - Lab report alerts
 - Payment confirmations
 - Urgent booking confirmations
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Feedback & Support

- Rate consultations
- Submit support tickets
- Access FAQs and clinic contact information